

Rajiv Aryal

Research-Oriented Web Systems and Security Professional

rajivary1@gmail.com | aryalrajiv.com.np | LinkedIn | GitHub | Medium

Research Profile

Computer science graduate and web systems practitioner with experience in usable security, authentication, infrastructure resilience, full-stack development, and digital research communication. Interested in research assistant opportunities involving web and software systems, cybersecurity, human-centered security, digital infrastructure, and responsible AI. Experienced in translating technical investigations into prototypes, documentation, and operational guidance for mission-driven organizations.

Education

Bachelor of Science in Computer Science and Information Technology

Tribhuvan University

Relevant coursework: Data Structures and Algorithms, Design and Analysis of Algorithms, Artificial Intelligence, Statistics, Mathematics, Software Engineering, Object-Oriented Programming, Databases, and Systems Analysis and Design.

Research Interests

Web and software systems; usable security and privacy; identity and authentication; organizational cybersecurity; digital research infrastructure; responsible and socially beneficial AI.

Technical Skills

Programming and web: Python, JavaScript, PHP, React.js, Next.js, Node.js, Express.js, WordPress, REST APIs, HTML, CSS, MongoDB

Security and infrastructure: OAuth 2.0, passkeys, DMARC, web application firewalls, CSP/HSTS and HTTP security headers, endpoint security, backup and disaster recovery

Research and development tools: Git/GitHub, Jupyter Notebook, NumPy, pandas, Matplotlib, Postman, Zotero

Selected Research and Technical Projects

Organizational Security and Backup Infrastructure Design

Google Workspace, passkeys, DMARC, enterprise browser management

- Investigated security architectures for authentication, credential management, browser administration, backup, and organizational identity systems.
- Implemented passkeys, DMARC, centralized browser policies, and recovery workflows; evaluated operational tradeoffs and infrastructure resilience.
- Produced technical documentation and operational frameworks for secure, scalable, and maintainable deployment.

Microsoft Login Plugin: Authentication Systems and WordPress Integration

PHP, OAuth 2.0, Microsoft Identity Platform, WordPress

- Investigated OAuth 2.0 workflows and Microsoft Identity Platform requirements for enterprise authentication in WordPress environments.
- Built a custom plugin implementing authentication, token handling, and API communication while examining interoperability and identity-management considerations.

Messenger Link Impersonation Vulnerability Research

GitHub

Node.js, Express.js, HTML/CSS, Nodemailer, responsible disclosure

- Investigated a potential impersonation vulnerability involving Messenger link previews through a controlled proof of concept focused on user trust and authentication risk.
- Documented the methodology and findings and submitted the issue through Meta's Bug Bounty Program in accordance with responsible-disclosure practices.

ImPower Healthcare Web Platform

Website

WordPress, PHP, JavaScript, CSS, responsive design

- Designed and implemented a mobile-first healthcare website from UI/UX planning through deployment, including custom functionality and third-party service integration.
- Evaluated analytics, responsiveness, and platform performance to improve usability across devices.

Professional Experience

Security Consultant — *We Animals*

Mar 2026–Jun 2026

- Designed and implemented organizational security infrastructure spanning passwordless authentication, credential management, endpoint protection, and disaster recovery.
- Deployed enterprise browser management, email authentication, and web application firewall controls; developed security playbooks, onboarding guides, and operating procedures.
- Delivered awareness sessions on phishing prevention, authentication, and backup and recovery practices.

Web Systems and Security Analyst — *ANDA*

Jan 2026–Apr 2026

- Analyzed and security-tested a Next.js and headless WordPress system, identifying application-security and performance gaps.
- Researched and implemented CSP, HSTS, and other HTTP security headers; documented the system to support maintainability and knowledge transfer.

Research Library Web Publisher — *Faunalytics*

May 2025–Dec 2025

- Transformed research reports into accurate, accessible, web-ready resources for an online research library.
- Reviewed, formatted, and published articles while improving consistency, content organization, usability, and accessibility with the Resource Library Manager.

WordPress and PHP Developer — *Mangobyte Digital*

May 2024–May 2025

- Developed reusable JavaScript, CSS, PHP, and WordPress components to improve maintainability and content-management workflows.
- Conducted technical audits, data extraction, and performance analysis; customized administrative systems to improve usability and functionality.

Website Manager — *Vegan Travel Asia, VegVoyages Foundation*

Jan 2023–Jan 2024

- Maintained accessible organizational websites and digital communication systems, resolving issues and optimizing platform reliability.
- Supported advocacy outreach through website operations and Google Ads management.

Web GIS Intern — *NAXA - Location Matters*

Apr 2022–Jul 2022

- Contributed to React.js front-end development and API integration; supported an Express.js and MongoDB back end and managed SEO metadata with React Helmet.

Teaching, Mentoring, and Leadership

Python Programming Trainer — *Krinjal Foundation / Techie Nepal / U.S. Embassy collaboration*

2021–2022

- Taught Python, computational thinking, and problem solving to high-school students; connected programming concepts to introductory Raspberry Pi projects.

Founding President — *Birendra Open Source Club*

2021–2022

- Co-founded an undergraduate open-source community, organized Git/GitHub and Hacktoberfest workshops, and mentored students in sustainable collaboration practices.

President — *CSIT Association of Nepal–Chitwan*

2019–2022

- Led teams delivering technical workshops, hackathons, conferences, and regional networking programs; coordinated logistics, outreach, and technical operations.

Fellowships, Selected Training, and Conferences

Worldbuilding AI Futures Course

Aug 2026

Completed the Worldbuilding AI Futures course.

EAGxIndia Conference

Sep 2026 (upcoming)

Planning to attend for AI safety discussions, professional networking, and exploration of collaboration and work opportunities in AI safety.

Futurekind AI Fellowship

2025–2026

Impact-focused training and project work on developing AI that benefits the planet and sentient life.

Future of AI, BlueDot Impact

2026

Studied AI and AGI development pathways, hands-on case studies, and societal implications.

OSM Hackfest, Chitwan Region

2023

Mentored teams in web development, technical problem solving, project preparation, and presentation.

CSITAN Hackathon

2022

Participated in the CSIT Association of Nepal's national hackathon.

Code Camp, Western Regional Campus

2019

Developed a mobile application designed to promote local tourism during the hackathon.